

2nd IMECHE UAS CHALLENGE
Critical DESIGN REVIEW



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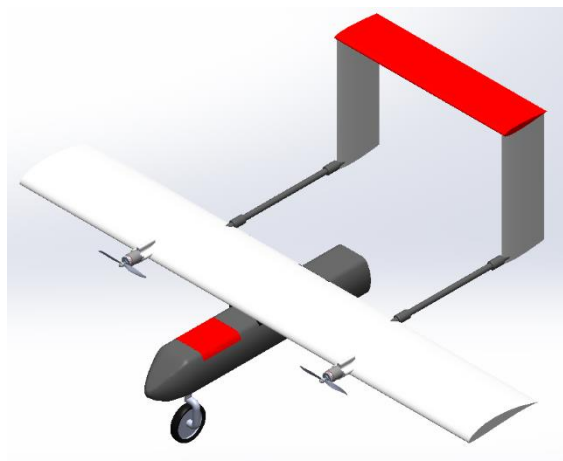


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SUMMARY DESCRIPTION OF DESIGN:

1. Description of proposed design:

The design of our Unmanned Aerial Vehicle (UAV) is guided by three core principles: manufacturing simplicity, top-tier performance, and cost-effectiveness. Every decision in the design process reflects these principles, resulting in a seamlessly crafted UAV. The emphasis on easy manufacturing ensures efficient production, while a dedication to performance excellence creates a UAV that maneuvers with remarkable skill.

Simultaneously, the commitment to cost-effectiveness grounds the design in practicality, ensuring each decision maintains economic feasibility without compromising functionality. This meticulous balance yields a UAV that epitomizes engineering precision, operational mastery, and cost-effectiveness. In summary, our UAV's design is a testament to creativity and pragmatism, delivering a dynamic and cost-effective solution that soars above expectations.

2. Changes since PDR:

- Our commitment to accuracy led us to revise the UAV weight from 6.4Kgs to 7.7Kgs, as initial estimates were rough approximations.
- Utilizing SolidWorks, we conducted meticulous calculations for UAV volume and airframe weight, triggering adjustments in other parameters due to the weight change.
- The semi-wing span has evolved from 2.4m to a refined 2.6m, enhancing the UAV's aerodynamic profile.
- Recognizing the need for optimization, the tail volume ratio has been adjusted from 0.55 to a more effective 0.6.
- In a strategic cost-saving move without compromising performance, we transitioned from the costly OS Motors OMA 5010/810KV to the budget-friendly EMAX GT2826/04 1090KV. Remarkably, both motors deliver the same thrust, ensuring efficiency without breaking the bank.

3. Contribution from sponsors:

There hasn't been any contribution from the sponsors yet.

4. Weight of cargo to be carried in main mission:

We transport approximately 1.7 kilograms of essential medical supplies to areas in need. The entire cargo is released in one go, with the UAV delivering a single payload at a time to serve the people in those affected regions.

5. Tasks to be undertaken in main mission:

The primary mission task is to efficiently deliver vital medical supplies to affected areas within the range of 5km. The UAV is optimized for accuracy and speed, ensuring timely and targeted deployment of essential supplies to those in need. A dropping mechanism has been designed for that purpose. It is equipped with a camera system with an On-board computer with a Machine learning Computer vision algorithm to detect the DZ and drop the payload at the location.

Project Management

The team has been steadily following that plan with minor changes in the timeline, with the main factors deciding it being the deadline of PDR, CDR and academic tests. The team is experienced in manufacturing of UAVs as we had taken part in the previous IMecHe UAS Challenge as well as the NDBFC'22, with successful other fixed-wing and rotary wing drones made before. Along with that the team is dedicated in its efforts often putting over-time to complete objectives before time. The team is however lacking in automation experience and thus more time was allocated to testing algorithms and autonomous controls on the UAV, especially with the challenges that come with stable take-off and landing of fixed-wing UAVs and the much more difficult image-detection task.

The plan starts from July where the Team was decided, and Team Leads were decided based on experience and team managing capabilities. They were divided into the

- **Design Team:** responsible for structural and mechanical design and analysis of the UAV to achieve the needed flight speed and patterns.
- **Automation Team:** responsible for Autonomous Flight, Controls, and making the path-finding and object detecting algorithm for the mission.

The team's collaborative efforts involve essential members like Ahmed Mohiuddin, who acts as a vital intermediary between the mechanical and design teams. Initially, the team invested the first two weeks in an in-depth literature review, exploring existing drone designs for disaster management. A critical decision-making point emerged in the choice between a high-plane for stability and a low-plane for increased maneuverability. Through basic simulations on SITL simulators, the team opted for a high-plane design, providing the required maneuverability with an enhanced turning curve while ensuring stability at speeds up to 20 kts. However, this decision led to a slight delay of one week in the overall design timeline. Concurrently, the Automation team delved into the intricacies of custom missions using the Python Dronekit API, concurrently integrating Raspberry Pi with the flight controller to enhance the drone's automation capabilities.

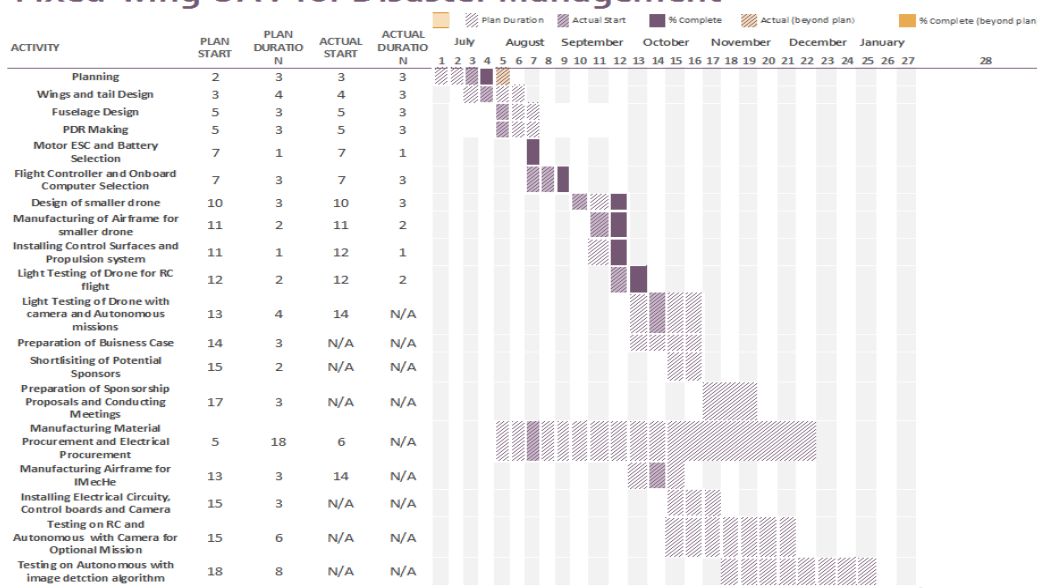
With students returning in September, a meticulous project timeline was devised, strategically allocating tasks until the end of October. This strategic alignment aimed to synchronize with midterms, allowing for efficient workload management. To facilitate experimentation and refinement, the team decided to construct a smaller trainer fixed-wing drone specifically tailored for automation mission testing. The team procured 3m width XPS foam from Lahore and other necessary components locally. A swift manufacturing process, utilizing a locally crafted hot wire cutter, enabled the team to construct the smaller drone within a span of two weeks. Subsequently, the Automation team initiated a series of comprehensive flight tests in October, yielding crucial insights into waypoint sensitivity, wind distortion effects, and camera angle considerations that might affect the project's overall success.

During the subsequent Critical Design Review (CDR) phase, the Automation team pivoted its focus towards intensive flight tests. Simultaneously, the design team and team lead commenced rigorous preparations for the CDR, emphasizing mutual review to ensure accuracy and completeness of their work. Dr. Kashif, our previous supervisor would leave during the semester and his supervision would continue till the mid of October till the CDR submission and thus, Dr. Sara Ali was contacted and would be the project supervisor for the future, granting the team access to a workspace in the Human Robotics Interaction Lab in SMME H-12 Islamabad. The construction plans involved dedicating three weeks to crafting the airframe, followed by a meticulous month-long process of integrating and calibrating the electrical components within the fuselage and wings. Anticipating potential delays in mid-November due to exams and team member engagements, the team adopted proactive scheduling measures to stay ahead of their established timeline.

Post-CDR, the team's focus will shift towards preparation for a comprehensive business presentation. Concurrently, the team will work with the sponsorship team to shortlist potential companies for partnerships and schedule meetings to secure sponsorships. This effort was planned to continue for a month, emphasizing thorough proposals and scheduled meetings. In the event of unsuccessful sponsorship attempts, the team committed to self-funding the remaining project requirements to ensure timely completion. Testing activities were rescheduled to occur post-Mid-Semester Exams, engaging 3-4 dedicated students in assessing flight conditions and evaluating the algorithms' effectiveness. The objective remained to conclude these tests before the end of December, aligning with students' departure for the semester break and minimizing the workload on team members engaged in concurrent semester projects.

Risks	Mitigation
Electrical Components not available locally	Parts to be ordered from abroad are pre-ordered once team is selected for fly-offs at the earliest note. However, most of them have already been procured by the team.
Team unskilled in Image detection and custom missions	Members of Automation team have taken courses from Deeplearning.ai to increase knowledge regarding ML and Computer vision. Similarly, Software-in-the-loop simulators are used to test custom mission on Raspberry pi and Mission planner.
Sponsors not found to fund drone.	Team will self-fund the drone to ensure that it is made.
Breakage of Electrical Components due to misuse	Cheap parts are ordered with replacement. expensive parts are handled with precautions such as using Pixhawk with shock pad and using Raspberry Pi with a static discharge resistant pad. Use of DMM and labeling of wires done to avoid short circuiting.
Machining of Accurate Airfoils and tails	Team has made its own sheets and XPS foam cutter and tested it to ensure accurate cuts. Inaccurate parts are sanded to ensure smoothness. We are also working on a CNC foam cutter to ensure 99.5% accuracy
Injuries during Manufacturing and Flight tests	All students have taken Occupational Health and Safety course in their university. Fire extinguisher and first aid kit kept in workspace and flight field. Pre-flight check list made keeping in mind safety and smooth operation. All members educated regarding Flight and Flights performed away from residential areas.

Fixed-wing UAV for Disaster Management



Mission Requirements Plan

Requirement	Verification
Complying with allowed airframe configurations	We have opted for Fixed Wing UAVs, a choice that complies with the given configurations in rulebook.
The Maximum Takeoff Mass (MTOM) shall not exceed 10 kg, which will comprise both the payload and UAV	The total Mass of UAV with payload and all the accessories included is 7.7 kg which lie under the given limit
The UAV shall be designed for rapid assembly/disassembly to fit into the Storage Container.	Implementation of removable landing gear and wings designed for easy detachment from the fuselage.
The UAS shall be designed to operate from within a 10m x 30m box, orientated within 30° of the wind direction. Landing includes touchdown and roll-out, with the UAS required to stop within the box.	The way the UAV takes off is carefully planned for a distance much shorter than the specified 30 meters.
The placement of the data logger should comply with the overall vehicle safety and should be fastened and secured properly to avoid any issue in flight and flight safety review.	We've strategically placed the data logger on the top of the fuselage to make sure it gets a good signal from the sky and stays safe during the entire flight.
The UAV design must incorporate a mechanism to attach the payload.	A dedicated storage compartment within the fuselage is provided for the payload, equipped with a precisely engineered payload deployment mechanism.
The limit of the maximum value of COTS components used is PKR 250,000	The total cost of COTS components is PKR 232,350 which lie under the limit of PKR 250,000.
Batteries used in the UA shall contain bright colors to facilitate their location in the event of a crash.	Battery will be covered with a reflective tape of bright color to be easily found in case of emergency

Propulsion:

Requirement	Verification
Nature of the proposed propulsion system for the UAV.	Electric propulsion system which is allowed in the rulebook.
Nature of UAV (fully autonomous, semiautonomous, or manual)	Fully autonomous takeoff and landing
The maximum Airspeed of the UA in level flight shall not exceed 60 KIAS	Design cruise speed is 15m/s which lies under the limit of 60 KIAS (30.86m/s)

Flight Controls

Requirements	Verification
Any radio equipment shall be compliant with PTA directives and licensed for use in Pakistan. It should have a reliable operating range of at least 1 km; control of the UAS and the FTS is 'Spread Spectrum' compliant to 100mW spread spectrum conforming to IR2030 and CE marked 4.	Complying with the PTA directives Telemetry operates at frequency of 433MHz while receiver at 2.4GHz. Both possess a 100mW spread spectrum, complying with IR2030 and bearing the CE mark.
Max allowed altitude is 100m.	The UAV is programmed to be operate at a max height of 20 meters.
The UA shall automatically return to the takeoff/landing zone or terminate the flight after a loss of data link of more than 30 seconds.	RTL (Return to Launch) mode will be initiated as programmed in the mission planner
Flight Termination commands for fixed-wing UA without an alternate recovery, such as a parachute.	If RTL is not activated due to any reason the flight termination system commands the UAV for immediate landing in the safe zone provided.

Navigation and Mission Requirements:

Requirements	Verification
To size the fuel/battery load, the design team should plan on a typical target Mission flight path with a distance of no more than 2-4 km from takeoff to landing.	Selection of battery based on maximum distance elucidated. ESC take 47A on full load while the Control boards take about 5.5 A. Using these we calculated that the 13000 mAh batteru will give us a contingency of 3-4 mins shall be added.
The UAS shall be designed to operate from within a 10m x 30m box, orientated within 30° of the wind direction. Landing includes touchdown and roll-out, with the UAS required to stop within the box.	The UAV is designed so that the take of distance is less than 30m using higher thrust to weight ratio for swift take-off.
- The Ground Control Station shall display the following information and be visible to the Operators, Flight Safety Officer, and Judges: 1. Current UA position on a moving map 2. Heading 3. Altitude (angles) 4. Battery Level/Consumption 5. Current Consumption 6. Local airspace, including any 'No-Fly' zones. 7. Search Area Boundaries. 8. Height AGL (QFE). 9. Indicated Airspeed (KTS).	All of these requirements by UAS are displayed by the mission planner of the ArduPilot on the GCS.  The screenshot shows the ArduPilot Mission Planner interface. On the left, there's a sidebar with various flight data: Altitude (9.99), Distance to WP (9.42), Time Elapsed (48.00), Time Remaining (278.21), Vertical Speed (0.00), and Altitude (169.82). The main area displays a map with a flight path and waypoints marked with green dots and yellow arrows. The top of the window shows 'ARDUPILOT' and some status icons.
Detecting A +/-X on the ground and placing a waypoint there and dropping the payload there.	Camera is installed to provide ground video from a height of maximum 25m. An image processing algorithm is run on the On-board Computer using the video feed and a Machine learning algorithm using Computer vision is used to detect the DZ. Once it is detected using DroneKit a waypoint is placed at that point.

1. Wing Design

1.1. Introduction

Following the preliminary design phase, extensive calculations have yielded three crucial parameters: Maximum Take-off Weight (MTOW) and initial estimates of Wing Loading (S_w). These parameters will form the cornerstone for the upcoming wing design phase. In this section, we will delve deeply into the intricate process of designing the wing, with a primary focus on defining the wing's geometry and meticulously choosing the most suitable airfoil profile to align with our precise requirements. These are sized keeping in view the design priorities and requirements stated in detail in the preliminary design review.

In Appendix A, we can find a flowchart for chronological order of design process (Sadraey).

At this stage it is important to note that a mathematical model in MS Excel has been provided to us by Dr. Douglas Hunsaker, the final iteration of which has been attached at the end of this report (See Appendix B) (Hunsaker, 2022). This model facilitates easy changes to our UAV design parameters and even uses MS Excel functions to optimize the design of the UAV.

1.2. Wing Location

Three of the most popular wing locations are: high, mid, and low wing. Each of this has its benefits and downfalls; therefore, through excesses literature review, a decision matrix has been charted up for the engineering analysis in Appendix D (Lennon, 1996) (Sadraey).

Keeping in view our design requirements, our wing will perform best in a high wing location as it is easy to fabricate and demonstrates good performance and stability. A mid wing is hard to manufacture, and a low wing is deficient in some areas too.

1.3. Airfoil Selection

Opting for the appropriate airfoil is a pivotal step in wing design, impacting both tail configuration and the overall aerodynamics of the aircraft. A decision matrix will help us pinpoint the optimal one for our drone's specific requirements (See table 2) (Michael S. Selig, 1995).

Table 2: Design Matrix for Airfoil Selection

Parameters and Weights	NACA-6412	MH-114	FX 63-137 Smoothed	E-214
Cd min (10%)	0.00906	0.00827	0.00711	0.00585
Cm (20%)	-0.1288	-0.1944	-0.2113	-0.1432
α_s (20%)	15.0	15.0	17.33	13.85
CI max (20%)	1.58	1.8034	1.9723	1.5471
(CI/Cd)max (20%)	142 @ 5.2	161.2 @ 3.5	151.544 @ 3.44	146 @ 4
Stall quality (10%)	gentle	gentle	moderate	Slightly sharp

For the airfoil selection, we analyzed 4 different airfoils on XFLR5 on the Reynolds number (1,026,000) expected at our cruise speed: NACA-6412, MH-114, FX 63-137, and E-214 (Atmospheric Properties Calculator, 2016). Out of these, FX 63-137 and MH-114 provided the necessary lift coefficient to match our geometry in general sizing (Michael S. Selig, 1995). Furthermore, among these two airfoils, MH-114 provided the highest performance, as can be observed in Appendix C, in terms of CI/CD vs Angle of Attack (AOA) and Cm vs AOA.

1.4. Aerodynamic Parameters

For a Fixed-Wing UAV, there are several aerodynamic parameters. For brevity, these are summarized in a convenient table with the reasoning for selecting the precise value.

Table 3: Aerodynamic Parameters and their reasoning

Aerodynamic Parameter	Value	Engineering Analysis and Reasoning
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Taper Ratio	1	The selection of the taper ratio primarily relies on achieving a specific lift distribution across the wing. Due to its lift distribution along the wings, an elliptical shape being considered ideal (Sadraey). Since it is difficult to manufacture and is costly at this stage, we have opted for a taper ratio of 1 (Sameer Bin Khalid, 2022).
Sweep Angle	0°	At speeds below Mach 0.3, the drawbacks caused by wing sweep are more significant than the benefits it offers. Sweep angle introduces significant manufacturing complexity which clashes our design priorities (Sameer Bin Khalid, 2022).
Dihedral Angle	0°	A dihedral angle is mainly used to increase the lateral stability of the aircraft. Since, we have used high-wing configuration, adding dihedral would be redundant.
Twist Angle	0°	A twist angle is used to delay the stall at the tip. Rotating the airfoil at the tip causes difficulties in manufacturing and its benefits do not outweigh the drawbacks.
Wing Incidence Angle	1°	An in-depth analysis was done on XFLR5 by importing the necessary dimensions and weights of the designed UAV. For a stable flight in cruise conditions, the wings must provide a Cl of 0.52 at the given wing area as calculated through the lift equation 15ms ⁻¹ . This corresponds to 1° incidence angle.

1.5. Use of Mathematical Model for General Sizing of Wings

As stated in section 1.1, the mathematical model attached in Appendix A provides us with general sizing of the UAV. The aerodynamic parameters, MTOW, and S_w are inserted in the fields filled green. Furthermore, the design requirements as detailed in preliminary design phase, such as the cruise speed and stall margin were inserted into the MS Excel Sheet. This provided us with initial iteration of the general sizing of the UAV. Subsequent iterations involved approximating the airframe weight increasingly accurately gave us with a more refined sizing of the wing.

1.5.1. Optimization of Aspect Ratio and Wing Loading

Aspect Ratio (AR) and Wing Loading were optimized using MS Excel's Solver Method. The exact command is attached in Appendix E. A summary of the method is detailed in table 4.

Table 4: Optimization of AR and S_w

	Parameter	Reasoning
Objective Function	Estimated Range	We have to maximize the estimated range.
Constraints	Semi-span	It must not be greater than 1.3m since the wing will start to sag.
	Aspect Ratio	We need our aspect ratio to be between 6 and 10.
	Wing Loading	It must be between 60 g/dm ² and 90 g/dm ² .
	Stall Onset	As explained in PDR, it must be 9 ms ⁻¹ .
	Cruise Speed	As explained in PDR, it must be 15 ms ⁻¹ .
	Root Chord	Due to the availability of material thickness of XPS foam, we need root chord to be between 0.4 m and 0.5 m.
	Weight	It is fixed at 7.733kg.
Decision Variables	Aspect Ratio	We need to adjust our decision variables to maximize our objective function given the constraints. AR = 6.3 and S_w = 71.6 g/dm ² in the final iteration.
	Wing Loading	

With this, our wing is completely defined. For a detailed description of wing parameters, Appendix B provides the necessary information.

2. Tail Design

2.1. Tail Configuration

The tail of the UAV could be designed in a multitude of ways. Appendix F provides a design matrix for different tail configurations for a comparative study. It is clear the Inverted U-tail is a perfect fit for our design requirements and priorities (Sameer Bin Khalid, 2022).

2.1.1. Horizontal Tail Design

The horizontal tail (HT) follows much of the same design as that of wings; however, the main purpose of the tail is control and stability rather than lift generation. Table 6 provides a detail on the HT parameters and their reasoning.

Table 6: Horizontal Tail Parameters and their Reasoning

Parameters	Value	Engineering Analysis and Reasoning
HT Volume Ratio	0.6	A tail volume ratio between 0.3 and 0.6 was recommended by Professor Hunsaker (Hunsaker, 2022). Through further literature review, we settled on a high tail volume ratio for increased control of the UAV (Lennon, 1996).
Moment Arm	0.9 m	The moment arm must be less than the semispan of the UAV (Lennon, 1996). A high moment arm would cause low HT area and vice versa. Therefore, at a modest addition to the weight of the UAV, 0.9 m generated minimal tail friction drag.
Aspect Ratio	4	A lower AR is desirable for the tail to minimize the bending load of the tail. Conversely, a high AR tail causes significant out of propwash region (Sadraey). Thus, a modest 4 was calculated by: $AR_{HT}=2/3(AR_w)$ (Sameer Bin Khalid, 2022).
Taper Ratio	1	For a Inverted U-Tail, a taper ratio is always 1.
Incidence Angle	-2°	An in-depth analysis was done on XFLR5 by importing the necessary dimensions and weights of the designed UAV. For a stable flight in cruise conditions, no pitching moment must be present. Therefore, the HT must be placed at a -2° AOA to counter the nose-down pitching moment.

2.1.2. Use of Mathematical Model for General Sizing of Horizontal Tail

The above parameters were filled in the fields colored green in the mathematical model sheet attached in Appendix B. The outputs of the mathematical model were the sizing parameters of the horizontal tail (See Appendix B).

2.1.3. Vertical Tail Design

Vertical Tail (VT) has been designed in the same manner as that of HT. Some parameters are detailed in table 7. These parameters were inputs to the mathematical model in Appendix B, which provided us with the general sizing of the VT.

Table 7: Vertical Tail Parameters and their Reasoning

Parameters	Value	Engineering Analysis and Reasoning
VT Volume Ratio	0.02	A tail volume ratio between 0.02 and 0.05 was recommended by Professor Douglas (Hunsaker, 2022). Through further literature review, we settled on a value of 0.02 for our 2 VTs (Lennon, 1996).
Semi-Span	0.25	Semi-span was dictated by the requirement that the HT's chord matched that of VT's to make a reasonable Inverted U-Tail. Therefore, a 0.25 m semi-span provided reasonable height to HT.

2.1.4. Airfoil Selection

For both the tails, NACA 0012 was selected since it is a symmetrical airfoil and widely used for the tail section in UAVs (Lennon, 1996). This provided us with a gentle stall which happens after the wings' airfoil -both of which are crucial for stability.

3. Fuselage Design

The fuselage is designed for flexibility, offering ample space for component placement and center of mass adjustments. It features a 110cm length, 19cm internal diameter, and a square cross-section for ease of construction. The fuselage, made from XPS foam, includes a front hatch for tool-free access to the component compartment and a transversely mounted main wing for landing gear placement.

4. Propulsion System

4.1. Power Requirements

For a plane which has to take-off and do some aerobatic maneuvers, the power requirements is between 110 Watts/kg and 152 Watts/kg (Guidelines to help you choose your Plane's Power System, 2023). Using this method, we need power ranging from 978.2W- 1351.7W. A single motor that produces this much power

would be too expensive and hard to procure readily. Therefore, we have decided to use 2 motor and propeller combinations as our propulsion system.

4.2. Motor and Propeller Selection

From a pool of locally available motors, a few are shortlisted in table 8 and a design matrix is constructed for engineering analysis and reasoning.

Parameters	EMAX GT2826/04 1090KV	T-Motor AS 2820 1250KV	OS Motors OMA 5010/810KV
Power Delivery (30%)	5	5	4
Weight (10%)	3	5	4
Cost (25%)	5	3.5	2.5
Propeller (15%)	4	4	3
Current Through (20%)	4.5	2	3
Result	4.55	3.88	3.28

The results from table 8 are clear. With EMAX GT2826/04 1090KV, a 10x5 propeller is recommended in the data sheet.

4.3. Propulsion Performance Simulation

The propulsion system is tested in a commercially available simulation software designed to test specifically UAV propulsion system: MotoCalc. Since the motor we used wasn't available in the motor suite of the software, we inserted our motor parameters, such as no-load current and resistance. (See Appendix G for complete information) The simulation was run and the aerodynamic opinion the software provided is as follows.

- The full-throttle motor current at the best lift-to-drag ratio airspeed (47.4A) falls approximately between the motor's maximum efficiency current (29.9A) and its current at theoretical maximum output (249.4A), thus making effective use of the motor.
- The static pitch speed (25.3m/s) is within the range of approximately 2.5 to 3 times the model's stall speed (9.3m/s), which is considered ideal for good performance.
- The static thrust (60.82N) to weight (75.86N) ratio is 0.80:1, which will result in very short take-off runs, no difficulty taking off from grass surfaces (assuming sufficiently large wheels), and steep climb-outs.

5. Payload Mechanism

The payload dropping mechanism largely remains the same as the PDR. Through the use of hook and strap belts easily available in the market, the payload can be secured with the dropping mechanism as it hangs as a separate entity. High performance servo would be required to bear such a heavy load; thus, metal servo with a maximum capacity of 1.8kgcm-1 were chosen for this job. When the dropping point approaches the cargo will be released via the servo.

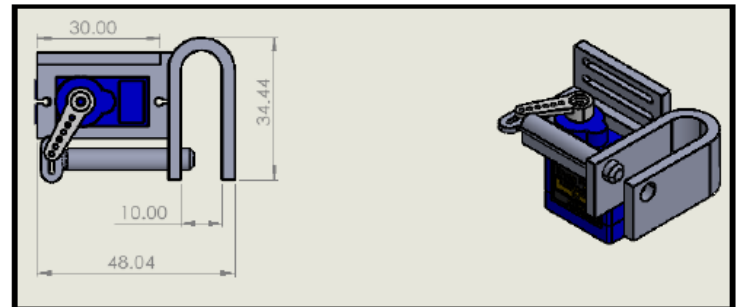


Figure 1: Payload Dropping Mechanism

6. Structural Performance of the UAV

6.1. FEA of Semi-Span of the Wing

We've opted to utilize XPS foam as the material, with the inclusion of two PVC pipes serving as spars. The analysis focuses on the left half of the wing and is subjected to a distributed load of 40 N. To simplify the analysis, we've chosen not to consider span-wise aerodynamic loads. It's important to emphasize that this FEA is exclusively centered on the wing and does not encompass other components of the airframe, once again for the sake of simplicity. The mesh aspect ratio is 2.3; therefore, it is of good quality.

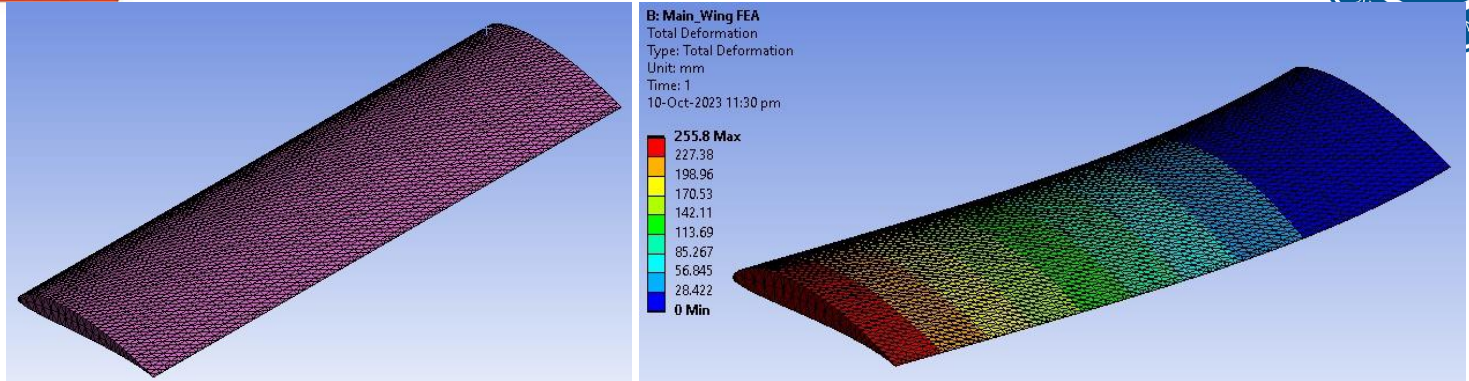


Figure 2: (Left) Mesh of Wings. (Right) Deformation in Wings. Note that for a 1.3m wing semi-span, 0.25m of deflection occurs under 40N load. The material doesn't yield yet.

6.2. FEA of Horizontal Tail

The same material (XPS) is used with no spars in the horizontal tail. A distributed load of 40N is applied. The same simplifications are applied, and good quality mesh is constructed.

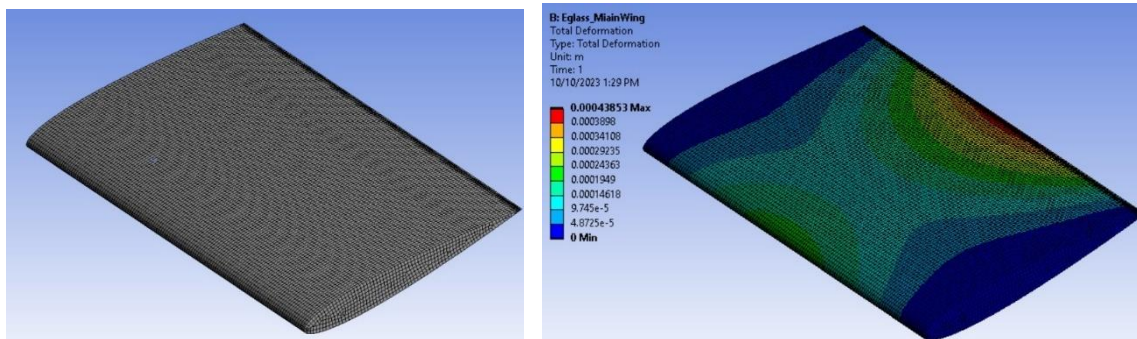


Figure 3: (Left) Mesh of Horizontal Tail. (Right) Deformation of Horizontal Tail.

6.3. FEA of Landing Gears

Structural performance of landing gears made from carbon fiber was also checked. They are attached in Appendix I.

7. Aerodynamic Performance

The Aerodynamic Performance of the UAV was estimated on XFLR5. The exact dimensions of the UAV which are stated in Appendix B were imported along with the plane inertia.

7.1. Coefficient of Lift vs AOA

As can be seen in figure, the aircraft produces approximately $C_L = 0.52$ at 0° AOA; therefore, the wing can lift the aircraft at cruise conditions. Also, at 10° AOA, the wing hasn't stalled yet which is a good indication, and the lift is increasing in a smooth gradient.

7.2. Pitching Moment vs AOA

It is observed in figure 5 that there is a nose up pitching moment of the aircraft at 0° AOA. However, this is kept deliberately since the motors are mounted on the wings, which will cause a nose-down pitching moment of approximately 2.6154 Nm given the C_g position in the z-direction is -0.05m. Therefore, the positive pitching up moment will counter the moment in cruise conditions. This positive pitching moment is done by placing the incidence angle of horizontal tail at precisely -2.0° .

7.3. C_L/C_d vs AOA

In figure 6, we can see that at the cruise condition, the UAV achieves almost the best lift-to-drag ratio, which is 17.024. This indicates that our aerodynamic parameters are very neat and the wings have very good aerodynamic performance

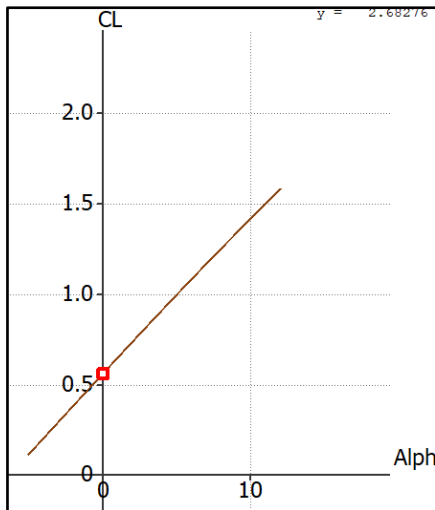


Figure 5: CL vs AOA

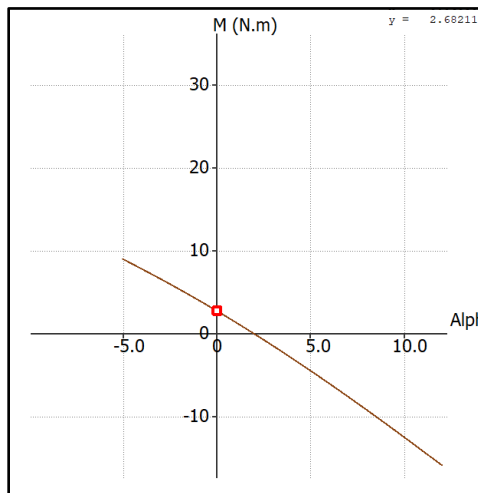


Figure 6: M vs AOA

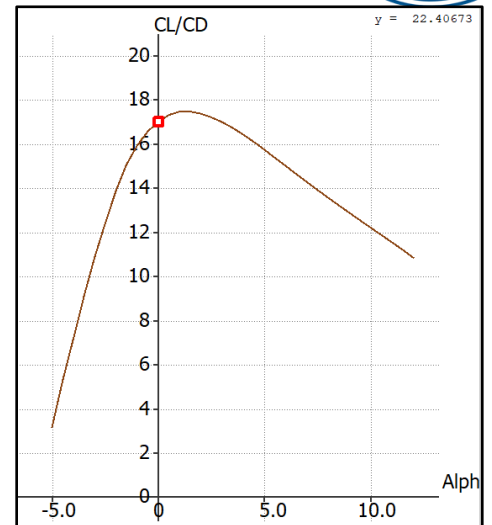


Figure 7: CL/Cd vs AOA

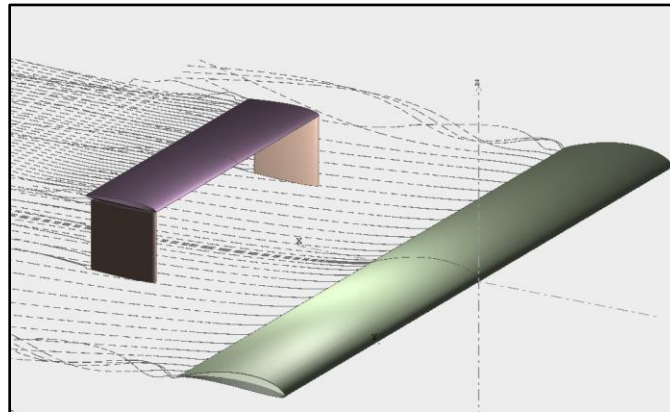


Figure 4: UAV Stream Flow in XFLR5

8. Weight Report

Item	Mass (g)	Item	Mass (g)
ESC 80A	126	Motors	470
OAK D	115	Propeller	46
Battery (3,000mAh 4s)	297	XT90 connector	51.2
Battery (10,000mAh 4s)	940	Miscellaneous wiring	70
Servo Motors	93.8	Insulation foam board	1800
Metal Servo	15	servo extension	68
Raspberry Pi	46	Motors	470
Main Landing Gear	500	Pixhawk	38
Nose Landing Gear	100	GPS Module	33
Receiver	15	Telemetry	63
Spar	1200	Parcel	1000
Total Weight (g)		7087	
Contingency Total (g)		7796	

8.1. Contingency Total

A 10% contingency total has been deliberately kept since many adhesives, such as glue gun, duct tape, and balsa planks are inevitably used to join or strength two surfaces. These cause the weight to be drastically more than estimated in the above table. Furthermore, the rulebook urges us to maintain a 10% contingency weight.

8.2. Location of Cg

The center of gravity will be located at 0.15m of the leading edge. This is due to the fact that the Cg must be located 10% of the Mean Aerodynamic Chord forward of the Neutral Point for aerodynamic stability (Aircraft Center of Gravity Calculator, 2023). For a complete overview, see Appendix H.

9. Flight Controller

The UAV is equipped with a Pixhawk 2.4.8 Flight Controller and operates with the PX4 autopilot software, specifically through Mission Planner. Its capabilities include autonomous takeoff, flight, mission execution, and landing. Additionally, it offers the functionality to automatically fine-tune the aircraft's PID values, enhancing its stability during autonomous flight. The selection of the Pixhawk 2 was driven by its 32-bit Cortex M4-Core Processor and good selection of embedded sensors such as the, The MPU6000 which is a high-performance 6-axis motion tracking device that combines a 3-axis gyroscope and a 3-axis accelerometer in a single package. This combined with the The ST Micro 16-bit gyroscope provides high accuracy and stability, while the ST Micro 14-bit accelerometer/compass provides precise heading information and the MEAS barometer The MEAS barometer is used to measure altitude and air pressure, which is essential for accurate altitude hold and altitude change control. All these sensors measure crucial data needed to learn the position and orientation of the drone to and external conditions needed to ensure smooth flight of the drone.

To establish communication with the aircraft, the Pixhawk 2 facilitates the transmission and reception of commands via telemetry connections. Our telemetry of choice is the 3DR 433MHz radio telemetry due to its operating range of 1.6 km which is enough for our mission. For GPS functionality, a GPS and Magnetometer module i.e., UBLOX NEO -M8N was chosen due to its accuracy of 0.6m, benefiting from being mounted atop the aircraft. It also had the QMC583L build-in digital compass with holder to easily mount on top of the UAV, along with having the necessary connection wire for Pixhawk

Power for the Pixhawk 2 is sourced from the aircraft's battery, with continuous voltage monitoring achieved through a dedicated power module. This voltage data is then transmitted to the ground control station via radio telemetry. Fail-safes are implemented to ensure the UAV's stability and operational efficiency. Our GCS of choice being Ardupilot due to its continuous display of the UAV and external parameters and ability to do stable autonomous missions with take-off and landing.

9.1. MISSION IMPLMENTATION:

With the Pixhawk in use, the mission needs to be uploaded into the Pixhawk's memory. The Mission Planner software allows for the creation of a mission by defining various waypoints. These waypoints specify the areas that need to be covered during the mission, and they also involve the use of servo mechanism to drop aid during disaster relief efforts.

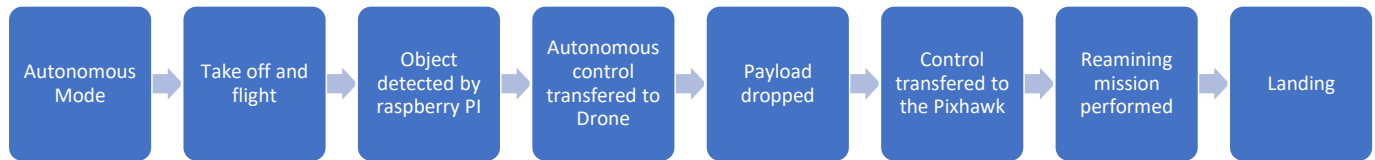
The mission will be executed entirely autonomously. In this mode, the Pixhawk will receive the mission data from the ground control system through the radio telemetry link, ensuring seamless coordination and control of the UAV during the mission.

The Pixhawk lacks the computational power and memory required for image manipulation. Consequently, we have opted to incorporate the Raspberry Pi 4B as a companion computer alongside the Pixhawk. The Raspberry Pi 4B 4GB with its Quad-core 1.5GHz Arm Cortex-A72 based processor functions as a



companion computer and communicates with the Pixhawk using the UART protocol. This setup allows for the necessary image processing capabilities, enhancing the UAV's functionality for tasks such as object detection during the mission.

9.2. THE COMPLETE PATH:



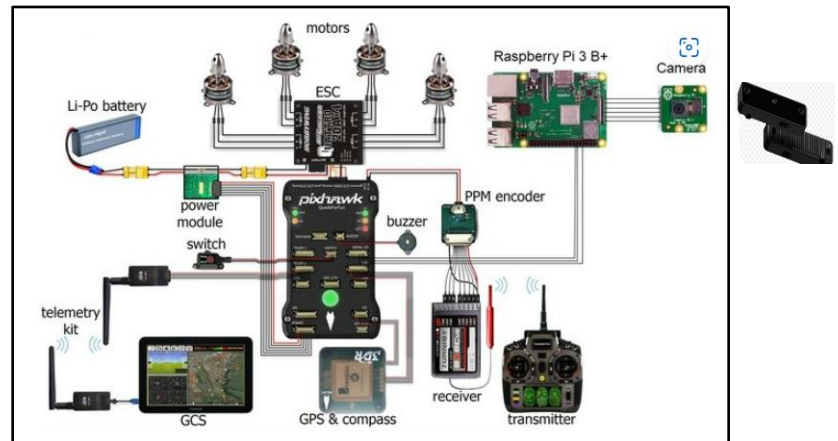
10. OAK-D FOR DETECTION:

An external camera, the OAK-D, is connected to the Raspberry Pi, enabling the Python software to perform object detection. The OAK-D camera is a powerful spatial AI tool that can concurrently run advanced neural networks. It offers in-depth information from two stereo cameras and captures color data from a single 4K camera in the center. The depth AI library serves as the interface for communication between the OAK-D device and the Python application running on the Raspberry Pi.

This camera plays a pivotal role in flag detection on the ground. When the camera identifies the +/X, it triggers the Raspberry Pi to initiate the mission execution process, thereby enabling the UAV to respond to the detected objects effectively.

11. Wiring Diagram:

The wiring diagram is similar to the one in the diagram however with 2 less motors and 6 servos connected across the Pixhawk servo rail for the 6 control surfaces. The rest of the diagram is the same.



12. Flight Termination System:

A Flight Termination System (FTS) is designed to initiate predefined actions when issues arise with the Unmanned Aerial Vehicle (UAV). This system integrates key components such as autopilot, Ground Control Station (GCS), and GPS. Specifically, the Pixhawk 2.4.8 autopilot's configurable failsafe feature is utilized, allowing customization via mission planner to execute programmed tasks under specific conditions.

- **Radio Failsafe:** When the pilot turns off the RC transmitter for over 0.5 seconds, the vehicle moves out of the RC range, loses signal for more than 0.5 seconds, or if there is a power loss in the receiver or a break in the connecting wires, the radio failsafe triggers. Depending on the situation, the UAV will either initiate Return to Launch (RTL) or bypass the failsafe in auto-mode.
- **Battery Failsafe:** This failsafe activates if the voltage falls below 22.8V or the remaining battery capacity is 10% or less of its original capacity. In either case, RTL mode is triggered to ensure a safe return of the UAV.
- **Return Home Signal (RTL Mode):** A designated channel in the transmitter-receiver setup allows the safety pilot to activate the Return to Launch mode, ensuring manual intervention capability.

- **Visual Line of Sight:** The UAV is equipped with a prominent visual marker on its exoskeleton, visible against the sky. Additionally, LEDs on the lower body enhance visibility in low-light conditions, aiding the flight operator's visual tracking.
- **Geofencing:** Geofencing functionality is enabled, ensuring that if the UAV attempts to leave the guided area, it switches to guided mode and returns to the predefined waypoint. It then loiters there, awaiting manual control takeover by the pilot.
- **GPS Loss:** The GPS health is continuously monitored, and if a GPS signal is lost for 3 seconds, the UAV enters Loiter mode for 30 seconds. If GPS does not recover within this timeframe, the UAV initiates RTL, ensuring a safe return home.
- **Ground Control Station Communication Loss:** The link between the Ground Control Station and the UAV is monitored through Mission Planner. If communication is lost for 10 seconds, the UAV will automatically trigger RTL, ensuring a secure response in case of communication breakdowns.

13. Image Preprocessing

Image preprocessing is essential to enhance the quality and reliability of target detection in images captured by the drone's camera. Many pre-processing steps can be included for better results. But for a trade off between accuracy and computation expenses, we can choose some steps from the following

Preprocessing Methods	Description	Implementation Algorithms	Priority
Thresholding	Convert the image to grayscale and apply thresholding techniques to segment the "plus symbol" from the background	Gray Scale Conversion	High Applying this step at last will make the target more visible in the image
Edge Detection	Apply edge detection algorithms to detect the edges of the "plus symbol."	Canny Sobel	Medium Can be a useful step, if there are enough resources to buy this

14. Target Detection

Once the image preprocessing is complete, we can proceed with target detection. There are various methods to detect objects in images, and the choice depends on specific requirements and the hardware power. Here's an overview of some common approaches:

Detection Method	Description	Requirements	Computation Expenses
Contour Detection	Filter the detected contours based on properties like area, aspect ratio, and orientation to identify the "plus symbol."	Relatively less training data needed, works well when the "plus sign" has distinct contours or shapes	Moderate
YOLO Algorithm	proposes using an end-to-end neural network that makes predictions of bounding boxes and class probabilities all at once	Requires a labelled dataset for training	High suitable for real-time applications

Faster R-CNN	A region based CNN, end-to-end, deep neural network	Requires labelled training data and more complex setup, offers high accuracy in object detection	High
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14.1. Target Detection

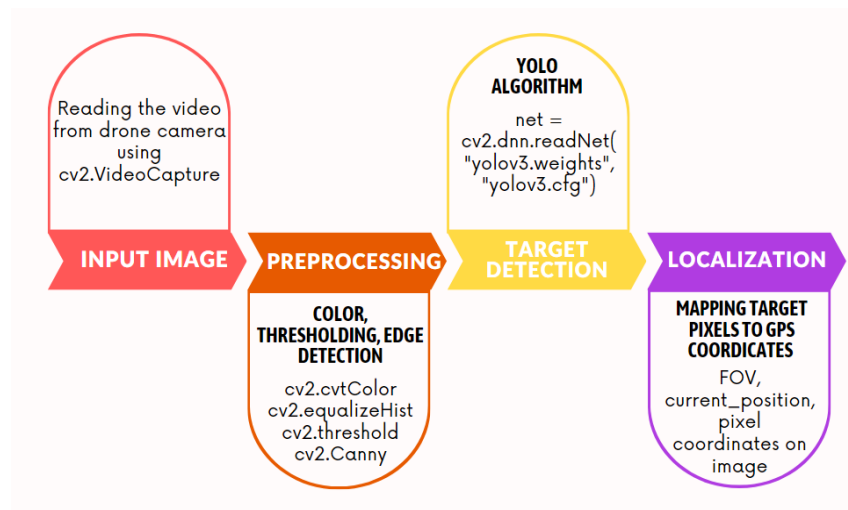
Considering that we know the Field of View (FOV), Pixel Coordinates of detected target in image, Drone GPS Location and Camera Parameters, We can calculate the target's GPS location mathematically

a. Calculate the horizontal angle (α) between the drone's optical axis and the line connecting the drone's camera to the detected target in radians: $\alpha = FOV * (2 * Px / ImageWidth - 1)$	b. Calculate the horizontal distance (D) from the drone to the detected target in meters using trigonometry: $D = (H/2) / \tan(FOV/2) * \tan(\alpha)$
c. Calculate the latitude (ΔLat) and longitude (ΔLon) offsets from the drone's current GPS location: $\Delta Lon = 0$ assuming camera is down $\Delta Lat = D / (earth.radius * \pi / 180)$ $\Delta Lon = 0$	d. Finally, calculate the GPS coordinates of the target: $Target_{Lat} = Lat_{drone} + \Delta Lat$ $Target_{Lon} = Lon_{drone} + \Delta Lon$

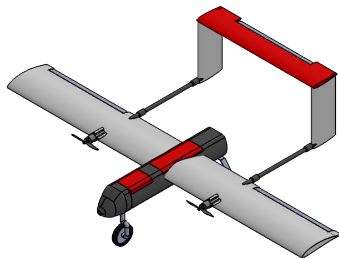
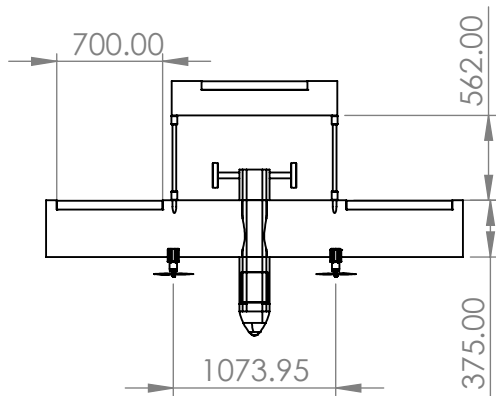
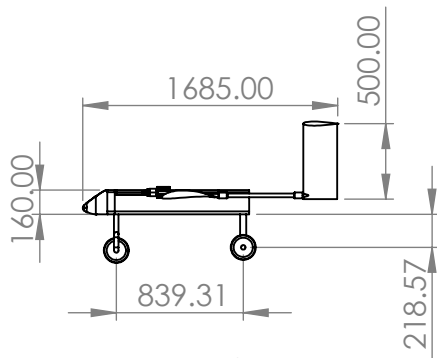
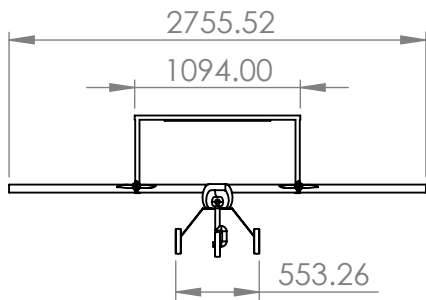
14.2. Algorithm and Preprocessing Selection Criteria

The selection of algorithms and preprocessing steps is a crucial aspect of the experimental process. The choice of algorithms should consider the available computational resources, the ease of target detection, the camera quality, and the required level of accuracy. Based on our current assumptions we will include Color correction, Thresholding and Edge detection in Preprocessing.

For the task of target detection, we priorities the use of either YOLO or Fast R-CNN. These choices align with our hardware capabilities, as they are optimized for efficiency, ease of training, and real-time performance. Moreover, through tests conducted on the drone they give promising results giving us an 80% success rate when flown from a height of 15 m and at 12-13 m/s.



Pipeline of the algorithm, cv2 refers to the python library opencv and yolo references the yolo algorithm.



SAFETY PLAN

1. DESIGN SAFETY FEATURES

S/No	Safety Feature	Description
1.	Reflective Strips	Strips of bright colors made of reflective tape for increased visibility in dim light, preventing accidents.
2.	Airframe Material	Extruded polystyrene (XPS) chosen for its strength (compressive: 250kPa, tensile: 450kPa, shear: 250kPa), protecting electronic components.
3.	Li-Po Battery Handling	Use of Li-Po Battery with 4S, with careful handling instructions provided, placed inside the fuselage and secured with Li-Po battery straps.
4.	Shock Absorbing Landing Gears	Landing gears mounted on the fuselage with shock absorbers to reduce vibrations and potential harm to the UAV.
5.	Modular Design	UAV designed modularly for easier and safer transportation, facilitating cost-effective part replacement in case of damage.
6.	Spare Components for Electric Systems	Spare components for electric systems, such as propellers and motors, carried to enable prompt repairs and mission continuation.
7.	Propulsion System Simulation	Simulation on software like MotoCalc and eCalc to determine motor currents (10A to 50A), ensuring compatibility with ESCs rated for 60A.
8.	Wires and Connectors Compatibility	Wires and connectors checked for compatibility; optimized power usage to prevent battery overload; insulated wires for safety and system integrity.
9.	Radio Signal Frequencies	Radio signal frequencies (2.4GHz for RC, 168MHz for FC, 433MHz for telemetry) within the safe range (100KHz to 300GHz), ensuring clear communication with Visual Line of Sight (VLOS).
10.	Payload Attachment	Payload attached to the fuselage at the center of gravity (COG) for stability during flight; ensures stability when the payload is dropped off.

2. OPERATIONAL SAFETY:

The following table summarizes the operational safety features, covering aspects from regulatory compliance and geo-fencing to the Flight Termination System (FTS) that ensures the safety of the UAV and its surroundings.

A	Requirements	Description
1.	VLOS Compliance	UAV must remain in Visual Line of Sight (VLOS) per FAA regulations. Reflecting tape, bright colors, and LEDs enhance visibility.
2.	Geo-fencing with Mission Planner	Geo-fencing Flight Termination System (FTS) set to 25m height and 2km radius using Mission Planner to ensure UAV stays within fixed boundaries.
3.	Remote Controller Connection	UAV always connected to remote controller for potential manual override. If manual override fails, FTS is activated. Handbook for safety and handling provided.
B	Flight Termination System (FTS)	Description
1.	Signal Transmission Loss	If transmitter loses signal transmission, drone returns to launch point for safety.
2.	Temperature Safety Threshold	Warning issued if component temperature exceeds safety levels; drone lands if temperature continues to rise beyond a threshold.
3.	Geo-fence Violation	If the drone enters a No-Fly Zone or exceeds geo-fence boundaries, it returns to the launch location.
4.	Battery Usage Monitoring	Drone initiates Return to Location if battery usage exceeds 90%, ensuring sufficient power for safe operations.

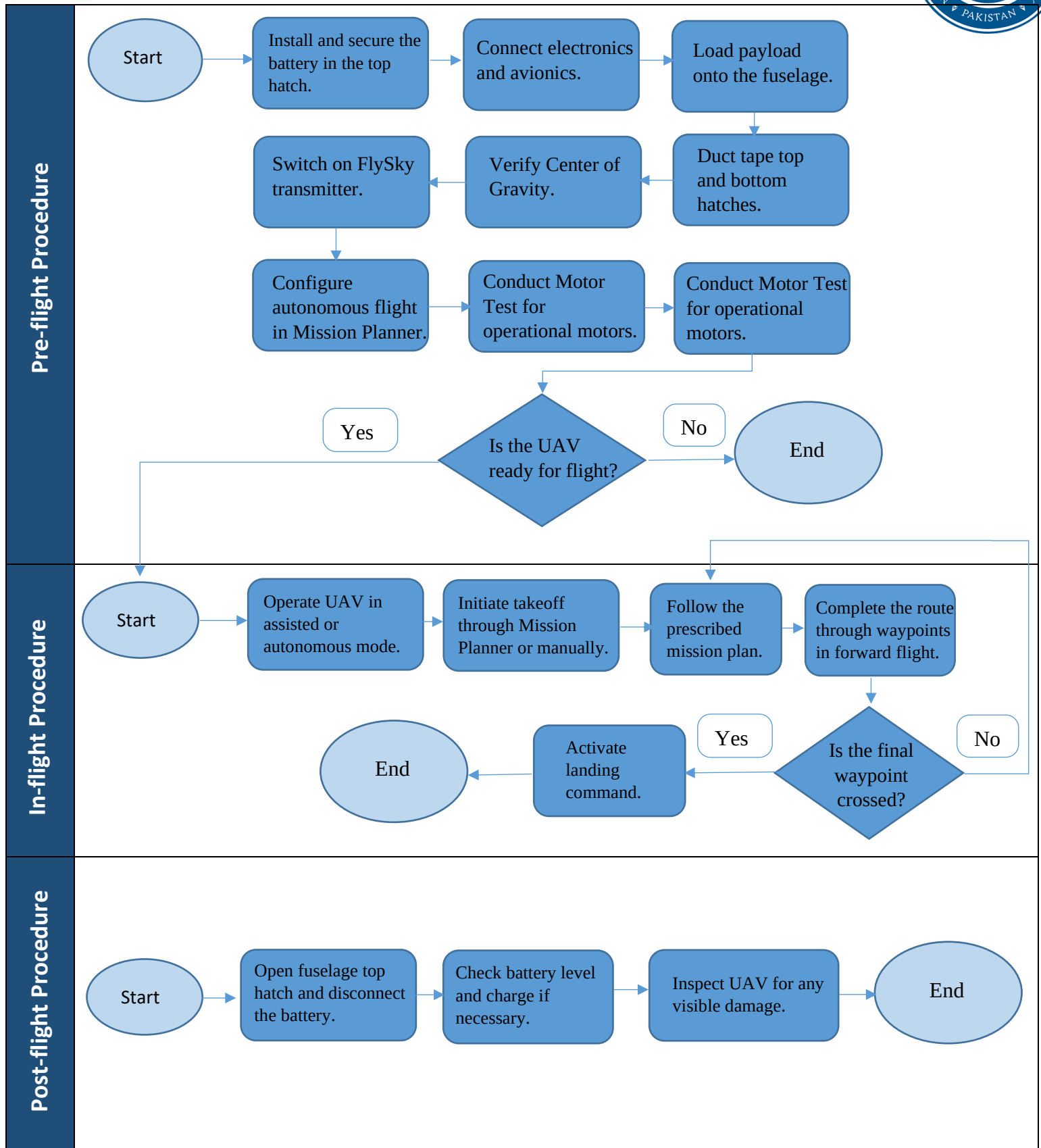
Emergency Procedures:

In UAV operations, having clear emergency procedures is crucial. If there's a loss of connection, backup plans ensure a pilot can regain control. Monitoring for low battery alerts prevents power-related issues. System malfunctions are addressed with protocols for quick fixes. Dealing with unexpected obstacles involves adaptive navigation to reduce risks. In emergency landings, specific protocols include stopping all UAV activities, initiating a controlled landing in a safe area, managing battery, and considering remote override. Post-incident investigation and analysis are conducted for improvement.

Table of Safety Risks and Their Mitigations:

Safety Hazards	Severity	Probability	Mitigation
Motor Failure	Marginal	Improbable	Good quality motors are used.
Battery Damage	Marginal	Improbable	Battery is placed in fuselage within allocated space and covered with Li-Po safety case.
Disconnected Equipment	Minor	Remote	Each equipment is to be double checked before flight.
ESC failure	Marginal	Improbable	Good quality ESCs are used with precise calculations on software.
Loss of Autonomy	Minor	Improbable	Flight tests are conducted and a pilot will take control.
Manual Override Failure	Minor	Improbable	Ensure receiver and transmitter are well Connected.
Extensive Vibrations	Minor	Occasional	Vibration Dampeners are used to avoid this problem.
Propeller Damage	Marginal	Remote	The drone will not strike or fall which may result in this.
Overheating /Flammability	Marginal	Remote	Thermal insulators are used. Battery is enclosed within Li-Po Safety case.
Bird Strikes	Marginal/Minor	Remote	Birds strike is still a problem and there is no clear solution.
Drone goes out of VLOS	Minor	Remote	Geo-fencing and UAV operator to make sure drone remains in VLOS
Equipment Damage	Marginal	Remote	Each equipment is enclosed within main frame.
Loose mechanical parts	Marginal	Remote	During assembly, it is made sure that all parts are well attached
Short Circuit	Marginal	Remote	All electrical components are connected correctly and well insulated.
Harmful frequency	Marginal	Improbable	The frequency of all components are within safe range of 30-300MHz
Loss of Connection	Minor	Remote	Failsafe have been set in case of Loss of connection

5. OPERATING PROCEDURES:



Manufacturing and support description

Initially research was done on typical materials used in manufacturing fixed wing UAVs and their availability. The processing required must be locally available and not be expensive.

1. Material selection

The material to be selected for the wings needs to be lightweight but durable. The UAV should also be manufactured using material which resists impact on crash, doesn't catch fire and avoids short circuits of the electronics.

- **Extruded Polystyrene Foam (XPS):**

We have selected high-density foam insulation board (usually made of expanded polystyrene or similar materials) that provides the required structural strength while remaining lightweight. We chose foam boards with a smooth surface to ensure better aerodynamics and ease of modification.

- **PVC Pipe:**

For spar of Main Wing and Tail booms we have chosen to use a $\frac{3}{4}$ inch PVC pipe that will provide structure to the main wing to make it capable to withstand the loads put on it during flight and to connect the tail to the main wing

2. Outsourcing :

Carbon fiber landing gear is to be outsourced online from AliExpress. The only outsourcing of major subsystems is for the electronic components, such as the motor, battery, controller, sensors, and communication devices. The outsourcing is justified by the fact that these components require specialized expertise, facilities, and certification that are beyond our scope and budget.

3. Manufacturing Techniques:

a. High intensity laser cutting:

For the wings and tail airfoil guiding plated were cut out using high intensity laser cutting. These guiding plates needed precise cutting as the airfoil must be precise and accurate, which was ensured by the CNC laser cutting.



b. Hotwire cutter:

The Hotwire cutter was made by the team from PVC pipe and some other material from a local hardware store. A 24V transformer was used to power the nichrome wire which being the heating element that did the cutting.



4. Wing/Tail construction :

- The cutting step involves using a hot wire cutter to cut the EPS foam into the desired shapes and sizes for the different parts of the UAV. This step requires careful measurements and calculations to ensure accuracy and precision.
- The shaping step involves using a sanding to smooth and refine the edges and surfaces of the foam parts. The shaping step requires attention to detail and quality control to ensure consistency and durability.
- We will use strong, lightweight adhesives or epoxies (Super Glue, Hot Glue, Two Part Epoxy) to securely join the foam components, such as the wings, elevator, and fin, to the fuselage.
- We will make sure each mating surface is clean and smooth to allow the Glue and mounting fixture to grip the components securely.
- Allow adequate drying and curing time for the adhesive to ensure a solid bond.

5. 3D printed motor mounts:

Motor mounts were designed on SolidWorks and 3D printed by a 3D printer owned by a team member.

6. Safety and environmental issues

The manufacturing process of the UAV is designed to be safe, efficient, and environmentally friendly. The safety issues include avoiding burns, cuts, and electric shocks from the hot wire cutter; avoiding dust inhalation and eye irritation from the sanding machine; and avoiding skin contact and respiratory problems from the spray gun. The safety measures include wearing gloves, goggles, masks, and protective clothing.

7. Storage and Transportation:

a. Storage:

- We will store the foam-insulated aircraft in a climate-controlled environment to prevent damage from extreme temperature and humidity changes.
- We will avoid stacking heavy objects on top of the aircraft, as foam can be susceptible to compression and deformation.

b. Transportation:

- When transporting the aircraft, we will place it in a protective container or custom foam cradle to prevent dents, dings, or warping during transit.
- We will secure the aircraft within the transportation container to prevent movement and vibrations that could damage the foam structure.

8. Innovative manufacturing aspects

To increase the strength of the wings while making it cheaper we used PVC spars which will increase the strength yet being very readily available and cheap. Hotwire cutters are not readily available in the market, so we used our expertise to design and manufacture one from readily available material including PVC pipes. CNC laser cutting was used to cut desired airfoil for wings tail and vertical stabilizer. Motor mounts were also designed, and 3D printed.

Qualification Test Plan

S.No	Objective	Method	Success criteria	Test results and date
1.	The speed must not exceed 60 knots (30.8m/s)	Experimental: During the flight test, the speed both cruise and stall can be obtained from the accelerometer.	Cruise Speed $\leq 15\text{m/s}$ Stall Speed $\leq 9\text{m/s}$	15m/s is the design speed. Experimental: Awaiting Manufacture
2.	MTOW less than 10 kg	Theoretical: The net weight is determined by combining the weight of each component as specified by the manufacturers with the weight of the frame from the CAD model. Experimental: Weighing Scales are used to find weight of UAV.	MTOW $\leq 10\text{ Kg}$	Theoretical: Weight report shows that MTOW of drone is 7.7 kg Experimental: Awaiting Manufacture
3.	2-4 km distance and battery sizing.	Theoretical: Battery of 13000 mah is being used that ensures that the UAV will cover the distance without the battery being used up. Experimental: The flight test is done to find the maximum range of UAV.	The UAV can cover the distance without totally consuming battery.	Theoretical: Wattage of Propulsion system is 700 Watts while for Control System is between 60-80 Watts. With a 13000 mAh battery we can expect a mixed flight time of 10.4 min which is enough to cover the mission
4.	Stress analysis of frame to check its durability	Theoretical: Analyses on Ansys. Stress analysis in ANSYS follows a structured process, starting with geometry and mesh generation, applying material properties, boundary conditions, and loads, followed by solving using the finite element method. Experimental: Loads will be added on the wings of the UAV to check if they can withstand it.	It can withstand the stress without breaking.	Simulation shows that our frame can withstand the applied stresses mentioned in design description. Experimental: Awaiting manufacture
5.	Optimal propulsion system	The thrust needed is evaluated by comparing it to the thrust generated by individual motors, which can be acquired from data sheets. To identify the ideal propulsion	Combined thrust of motor and lift generated by wings should be more than 7.7 kg at 50% throttle	MotoCalc calculation done which resulted in a flight time of 10.4 min with excess thrust left by motor during mixed flight Experimental:

		system for the UAV, conduct tests on different systems, assessing factors like flight duration, energy consumption, payload handling, and reliability. Analyze the data to determine the most efficient and suitable propulsion system that meets the UAV's mission requirements.		Awaiting manufacture
6.	Fulfillment of all the Requirements by GCS	The features of GCS are compared with those of Mission Planner.	It should display all the parameters that are mentioned in the rulebook.	ArduPilot with Mission Planner shows all the necessary parameters mentioned in the Rulebook
7.	COTS items $\leq$ 250,000 PKR	The cost of each component is added.	COTS items $\leq$ 250,000 PKR	Cost of COTS items is 232,350 PKR with shipping.
8.	Geofencing	Set up the geofence settings within the aircraft's avionics system, specifying the borders, altitude constraints, and any pertinent parameters requiring examination. Chart a flight path that brings the aircraft in proximity to or across the established geofence boundaries.	The UAV shall remain within geofence of 1km zone with maximum altitude of 100m.	Geofencing done by Mission Planner and airplane configured to not go out of range.
9.	The UAV should hold up in all weather conditions	Flight tests will be done in harsh conditions. Begin with test flights in favorable weather to set performance benchmarks. Then, expose the UAV to rainy and wind conditions to evaluate its performance, checking for any negative impacts on propulsion, sensors and ensuring that it maintains its intended flight path.	The UAV should be designed to operate in winds of up to 20 kts gusting up to 25 kts, and light rain. The UAS shall typically be capable of take-off and landing in crosswind components to the runway of 5 kts with gusts of 8 kts.	XPS Board is hydrophobic and fire-resistant thus allows no impedance in electrical system. Otherwise, Electrical sub-system is properly insulated with vents to allow cooling and prevention of short circuit. <u>Experimental:</u> Awaiting manufacture

Cost Breakdown

COMPONENTS	QUANTITY	COST (PKR)	TOTAL COST(PKR)
ESC 60A	2	Rs 5,400.00	Rs 10,800.00
OAK D Camera	1	Rs 55,000.00	Rs 55,000.00
Battery (3,000maH 4s)	1	Rs 6,500.00	Rs 6,500.00
Battery (10,000maH 4s)	1	Rs 24,500.00	Rs 24,500.00
Servo linkage	4	Rs 40.00	Rs 160.00
Push Rod	4	Rs 30.00	Rs 120.00
Horns	4	Rs 55.00	Rs 220.00
9g Servo Motors	5	Rs 380.00	Rs 1,900.00
Payload Servo	1	Rs 1,850.00	Rs 1,850.00
Motors (OMA 5010 810KV KV)	2	Rs 13,500.00	Rs 27,000.00
Propeller(13x8)	2	Rs 1,150.00	Rs 2,300.00
XT60 connector	4	Rs 200.00	Rs 800.00
Miscellaneous wiring	2	Rs 200.00	Rs 400.00
PVC Tail Booms	2	Rs 500.00	Rs 1,000.00
Vinyl Sheets	6	Rs 250.00	Rs 1,500.00
Foam Core	3	Rs 2,214.00	Rs 6,642.00
servo extension	4	Rs 160.00	Rs 640.00
Main landing gear Wheels	2	Rs 280.00	Rs 560.00
Main Landing Gear	1	Rs 6,700.00	Rs 6,700.00
Nose landing gear wheels	1	Rs 280.00	Rs 280.00
Nose landing gear	1	Rs 2,200.00	Rs 2,200.00
Nails/Bolts/Wiring	1	Rs 500.00	Rs 500.00
Battery Strap	1	Rs 500.00	Rs 500.00
Transmitter/Receiver	1	Rs 16,500.00	Rs 16,500.00
Pixhawk	1	Rs 28,500.00	Rs 28,500.00
GPS Module	1	Rs 4,800.00	Rs 4,800.00
Telemetry	1	Rs 10,000.00	Rs 10,000.00
Shock Absorber Mount	1	Rs 800.00	Rs 800.00
Double tape	1	Rs 120.00	Rs 120.00
Epoxy/Glue	1	Rs 1,000.00	Rs 1,000.00
Raspberry Pi 4B	1	Rs 45,000.00	Rs 45,000.00
Manufacturing Equipment	1	Rs 2,000.00	Rs 2,000.00
Total			Rs 260,792.00
Contingency Total			Rs 286,871.20

Cost of COTS components.

COMPONENTS	QUANTITY	COST (PKR)	TOTAL COST(PKR)
ESC 60A	2	Rs 5,400.00	Rs 10,800.00
OAK D Camera	1	Rs 55,000.00	Rs 55,000.00
Battery (3,000mAh 4s)	1	Rs 6,500.00	Rs 6,500.00
Battery (10,000mAh 4s)	1	Rs 24,500.00	Rs 24,500.00
9g Servo Motors	5	Rs 380.00	Rs 1,900.00
Payload Servo	1	Rs 1,850.00	Rs 1,850.00
Motors (OMA 5010 810KV KV)	2	Rs 13,500.00	Rs 27,000.00
Transmitter/Receiver	1	Rs 16,500.00	Rs 16,500.00
Pixhawk	1	Rs 28,500.00	Rs 28,500.00
GPS Module	1	Rs 4,800.00	Rs 4,800.00
Telemetry	1	Rs 10,000.00	Rs 10,000.00
Raspberry Pi 4B	1	Rs 45,000.00	Rs 45,000.00
Total			Rs 232,350.00

Cost breakdown by Category

Breakdown by Category	
Manufacturing	Rs 22,502.00
Avionics	Rs 139,090.00
Image Detection	Rs 100,000.00

Cost breakdown by Funding

Breakdown by Funding	
Already Procured	Rs 86,900.00
Sponsored	Rs 55,000.00
To Be Purchased	Rs 118,892.00

Avionics:

Majority of the electronic components including the Battery, Motors, Servos, Pixhawk, and other Miscellaneous wire etc are available (Electrobes) located in Islamabad and those components not available will be ordered for arttechhobbies.com and smarthobby.pk. The cost added are from the websites (electrobes.com, arttechhobbies.com and smarthobby.pk)

The telemetry which usually costs PKR 20,500 was bought for PKR 10,000 from a seller in Karachi who sells used electronics (it was made sure before buying that it properly worked.)

The Transmitter and Receiver Pair were already owned by one of the team members.

Image Detection:

OAK D Camera which costs about PKR 55,000 was sponsored by Aerial Robotics Lab in SINES (School of Interdisciplinary Engineering & Sciences for the image detection. Raspberry Pi 4B 4GB was previously owned by team lead.

Manufacturing Materials

Foam core was ordered from a factory in Lahore through a dealer in Rawalpindi.

Landing gear was previously owned by the team ordered from Aliexpress.com. Other manufacturing materials are to be bought from the local hardware store.

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Appendix A

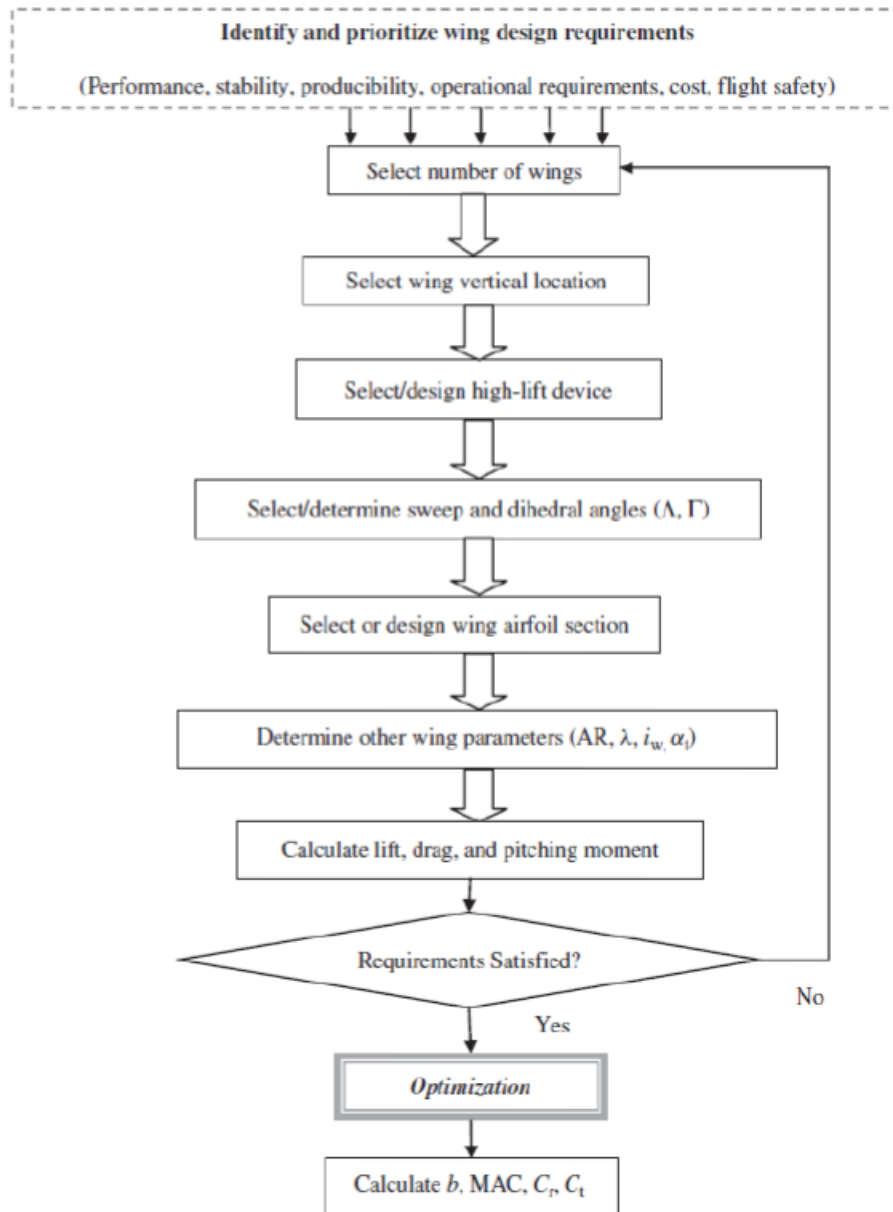


Figure 31. Design flowchart of the Wing [10]

[Edit](#)

Appendix B

Mission Requirements	
Endurance	10 min
Range	10 km
Stall Margin	6 m/s
Min Airspeed	9 m/s

Payload	
Payload Weight	3.106 kg
Payload Power	50 W

Aircraft	
Airframe Weight	3.3 kg
Avionics Power	500 W
Fuselage Drag Coeff.	0.03
Zero Lift Drag Coeff.	0.015
Air Density	1.204 kg/m ³

Propulsion/Batteries	
Battery Pack Capacity	13000 mAh
Energy Density	145 W-h/kg
Cell Voltage	3.7 V
Cells in Series (#s)	4
Number of batteries	1
Total Battery Capacity	13000 mAh
Voltage	14.8 V
Batteries Weight	1.327 kg
Total Capacity Used	80 %
Overall efficiency	40 %
Propulsive Power	335.05 W
Cruise Current Draw	22.64 A

Main Wing	
Wing Loading	71.6 g/dm ²
Aspect Ratio	6.3
Taper Ratio	1
Oswald Efficiency	0.85
Wing Area	1.080 m ²
Semispan	1.300 m
Root Chord	0.415 m
Tip Chord	0.415 m
Average Chord	0.415 m
Max Lift Coefficient	1.44
Airfoil	mh114
Horizontal Stabilizer	
Volume Ratio (0.3 - 0.6)	0.6
Moment Arm	0.9 m
Aspect Ratio	4
Taper Ratio	1
Area	0.299 m ²
Semispan	0.547 m
Straight TE Sweep	0.00 deg
Root Chord	0.273 m
Tip Chord	0.273 m

Vertical Stabilizer	
Volume Ratio (0.02 - 0.05)	0.02
Moment Arm	0.9 m
Semispan	0.25 m
Taper Ratio	1
Area	0.062 m ²
Root Chord	0.250 m
Tip Chord	0.250 m

Appendix B

[See Equation Sheet Here](#)

General Results			
Total Mass	7.733 kg		
Mass Fraction	57.33 %		
Min Drag Airspeed	11.60 m/s	41.75 km/h	25.94 mph
Min Power Airspeed	8.81 m/s	31.72 km/h	19.71 mph
Min Airspeed (Stall Onset)	9.00 m/s	32.40 km/h	20.13 mph
Reynolds number	419092.7	4.1909 x10 ⁵	
Condition Results			
Cruise Airspeed	15 m/s	54.00 km/h	33.55 mph
Lift Coefficient	0.52	Lift	7.73 kg
Drag Coefficient	0.0611	Drag	0.91 kg
Lift to Drag Ratio (L/D)	8.49		
Stall Margin	6.00 m/s	21.60 km/h	13.42 mph
Estimated Flight Time	10.4 min	0.17 hr	
Estimated Range	9.4 km	5.83 mi	

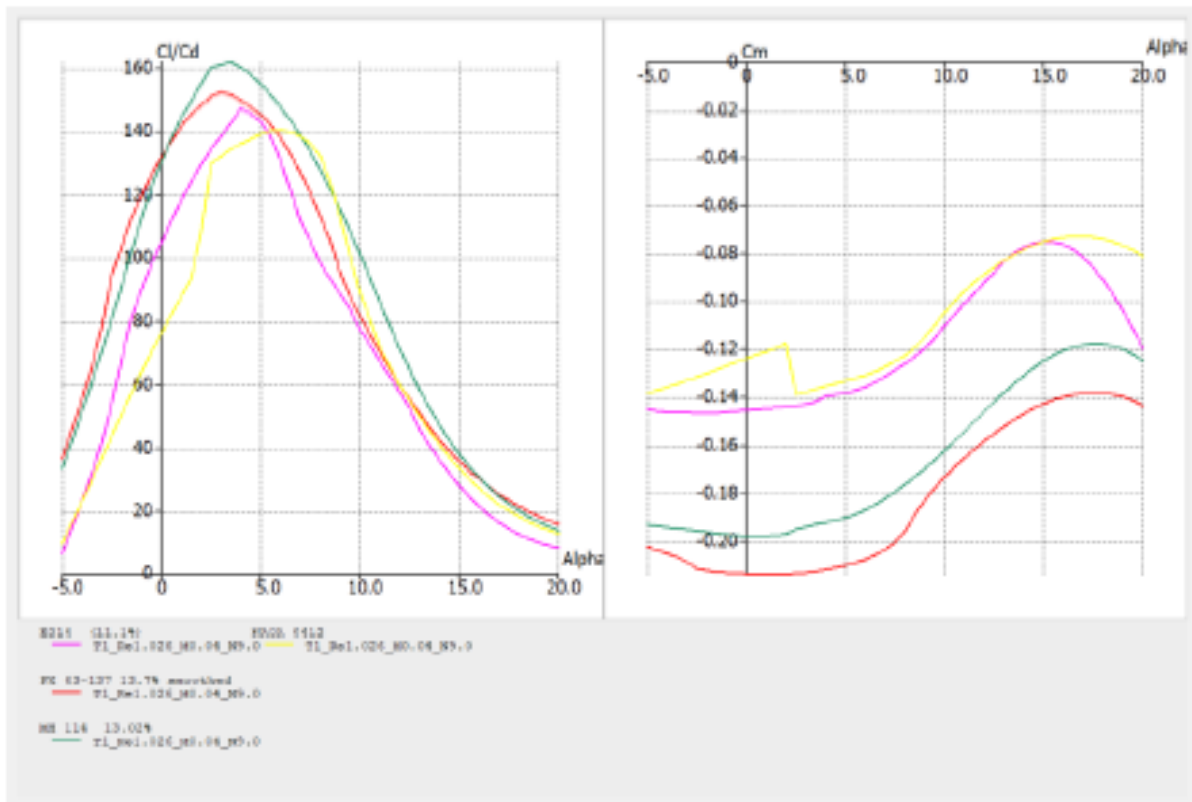
Typical Wing Loading	
Glider	31 g/dm ²
Trainer	46 g/dm ²
Sport Plane	62 g/dm ²
Fighters	76 g/dm ²

main wing Swe	0 deg
---------------	-------

Mass Fractions	
Composite	50 - 90 %
Balsa	35 - 50 %
Foam	< 35 %

Cell Voltages	
Lithium Polyme	3.7 V
Lithium Ion	3.6 V

Appendix C



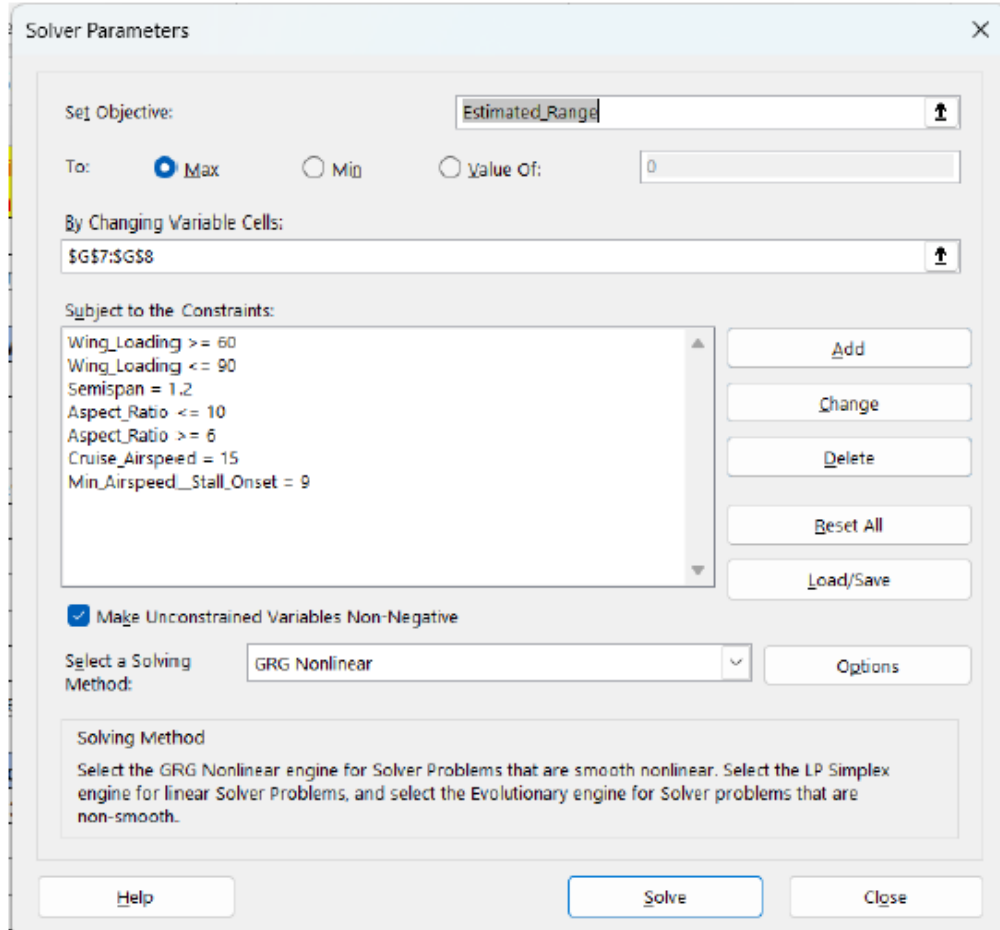
Appendix D

Parameters	High	Mid	Low
Weight (15%)	4	5	4
Stall Speed (15%)	5	2	3
Cost (20%)	4	5	4
Ease of Manufacturing (20%)	5	2	4
Longitudinal Stability (15%)	3	4	3
Lateral Stability (15%)	4	2	3
Result	4.2	3.35	3.55

Figure 1: Design Matrix for Wing Location.

Appendix E

Using Excel's Solver Method to maximize "Estimated Range" keeping in view the required constraints as detailed in the section "Design Requirements".



The image shows the "Solver Parameters" dialog box in Microsoft Excel. The "Set Objective:" field contains "Estimated_Range". The "To:" section has three radio buttons: "Max" (selected), "Min", and "Value Of:", with a text box containing "0". The "By Changing Variable Cells:" field contains "\$G\$7:\$G\$8". The "Subject to the Constraints:" list contains six constraints: "Wing_Loading >= 60", "Wing_Loading <= 90", "Semispan = 1.2", "Aspect_Ratio <= 10", "Aspect_Ratio >= 6", and "Cruise_Airspeed = 15". Below these is "Min_Airspeed_Stall_Onset = 9". To the right of the list are buttons for "Add", "Change", "Delete", "Reset All", and "Load/Save". Below the constraints list is a checked checkbox "Make Unconstrained Variables Non-Negative". The "Select a Solving Method:" dropdown is set to "GRG Nonlinear", with an "Options" button next to it. A "Solving Method" section at the bottom provides instructions on selecting the GRG Nonlinear, LP Simplex, or Evolutionary engine. At the bottom of the dialog are "Help", "Solve", and "Close" buttons.

Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

- Wing_Loading >= 60
- Wing_Loading <= 90
- Semispan = 1.2
- Aspect_Ratio <= 10
- Aspect_Ratio >= 6
- Cruise_Airspeed = 15
- Min_Airspeed_Stall_Onset = 9

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method: Options

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Help Solve Close

Appendix F

Table 5: Tail Configuration Design Matrix

Parameters	Conventional	V-tail	Inverted U-tail
Control (30%)	3.5	3	5
Stability (20%)	4	3	4
Drag (10%)	4	5	3
Ease of Manufacturing (25%)	4	3	3.5
Weight (15%)	3.5	5	3
Result	3.78	3.5	3.93

Appendix G

MotoCalc 8.09 Workbench

Project Edit Motor Battery Filter Drive System Speed Control Airframe Options Update Help

Motor

Name: EMAX 2826

Motor Constant: 1090 rpm/V

No-load Current: 2.5 A Design

Resistance: 0.045 Ω Tests

Weight: 6.173 oz Catalog

☒ Brushless ☒ Out-runner

New Open Save

Battery

Cell: Tattu C

Cell Capacity: 10000 mAh @ 3.7 V

Impedance: 0.004 Ω Chemistry: LiPo

Cell Weight: 8.29 oz

Series Cells: 4 to 4

Parallel Cells: 1 to 1

New Open Save

Filter

Name: Astro Cobalt 035, 05, 25, 40, Spc

Maximum Current: 30 A

Maximum Loss: W

Min Motor Efficiency: %

Max Motor RPM: rpm

Minimum Thrust: oz

☐ Use It New Open Save

Drive System

Description: Astro 035-15 2.38:1 Gearbox

Gear Ratio: 1 :1 to 1 :1 by 1 :1

G.B. Efficiency: 93 % Weight: 0.81 oz

Propeller Diam: 10 to 10 by 10 in

Propeller Pitch: 5 to 5 by 5 in

Const... P.Const: 1.17 Num Blades: 2

T.Const: 1 Num Props: 2

Propeller Ducted Fan

New Open Save

Series Motors:

Parallel Motors: 2

Speed Control

Name: Astro 210

Resistance: 0.003 Ω

Maximum Current: 60 A

Weight: 2.222 oz

☒ Brushless ☒ High-rate

Number of ESCs: 2

New Open Save

Airframe

Name: Great Planes Spectra

Wing Span: 102.36 in

Wing Area: 674.00 sq.in

Empty Weight: 236.34 oz

Coeff... Clopt=0.97
Climax=1.94
Cd=0.117
Cd=0.015

New Open Save

Compute Report...
Close
Help...

Appendix H

Aircraft Center of Gravity Calculator

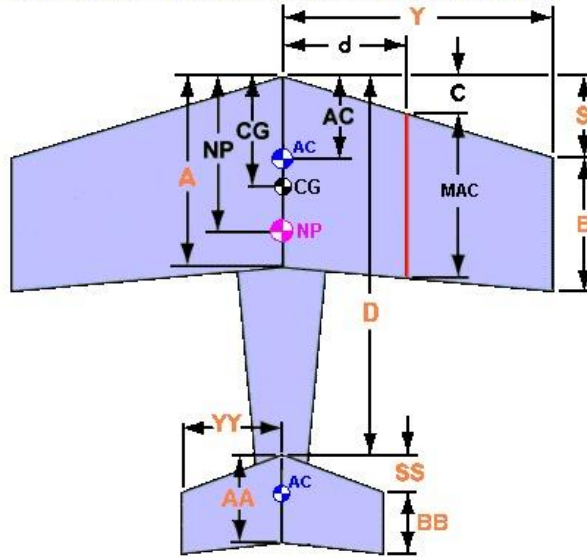
Aerodynamic Center (AC), Mean Aerodynamic Chord (MAC), Center of Gravity (CG), Neutral Point (NP) and Wing Area

Wing Root Chord (A):	0.415
Wing Tip Chord (B):	0.415
Wing Sweep Distance (S):	0
Wing Half Span (Y):	1.3
Stabilizer Root Chord (AA):	0.273
Stabilizer Tip Chord (BB):	0.273
Stabilizer Sweep Distance (SS):	0
Stabilizer Half Span (YY):	0.547
Distance between both LE's (D):	0.953
Stabilizer Efficiency [*] :	Std. ▼
Enter Static Margin, then	Click 10 %
Mean Aerodynamic Chord MAC =	0.42
Sweep Distance at MAC (C) =	0
From Root Chord to MAC (d) =	0.65
From Wing Root LE to AC =	0.1
From Wing Root LE to NP =	0.2
From Wing Root LE to CG =	0.15
Wing Area =	1.1
Stabilizer Area =	0.3
Wing Aspect Ratio =	6.27
Tail Volume Ratio, Vbar =	0.61

Enter the variables at left using the same units for all entries.

For an aircraft to be stable in pitch, its CG must be forward of the Neutral Point NP by a safety factor called the **Static Margin**, which is a percentage of the MAC (Mean Aerodynamic Chord).

Static Margin should be between 5% and 15% for a good stability.



Low Static Margin gives less static stability but greater elevator authority, whereas a higher Static Margin results in greater static stability but reduces elevator authority.

Too much Static Margin makes the aircraft nose-heavy, which may result in elevator stall at take-off and/or landing.

Whereas a low Static Margin makes the aircraft tail-heavy and susceptible to stall at low speed, e. g. during the landing approach.

Appendix I

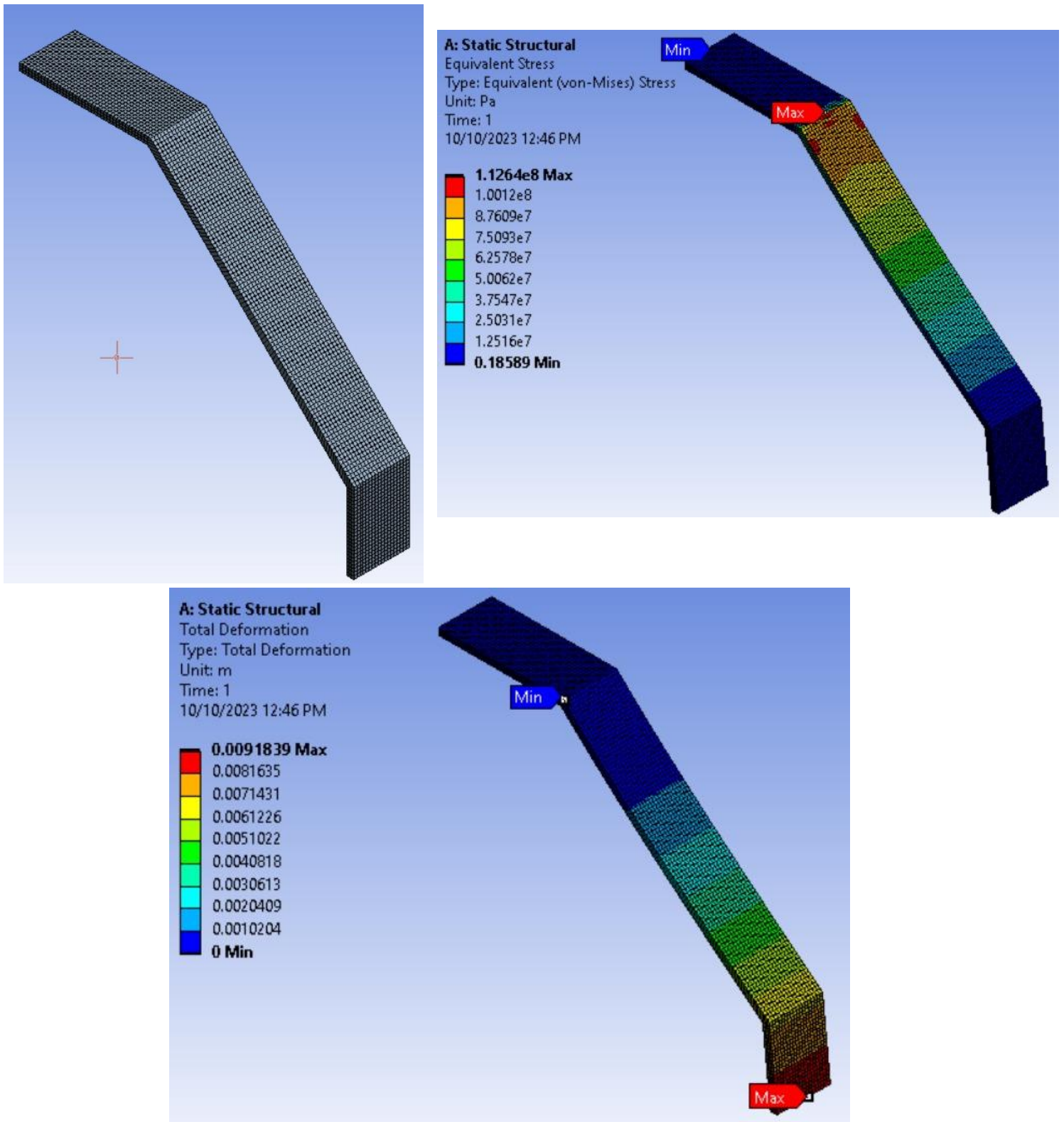


Figure 1: (Top Left) Mesh of Landing Gears. (Top Right) Stress of Landing Gears. (Bottom) Deformation of Landing Gears